HW4 Question 1

```
clear; clc;

f = @(x,y) 1./(3*(y.^2));
y_exact = @(x) x.^(1/3);          % Part a

h = -0.05;                          % backward step (1 -> 0) with |h|=0.05
x_grid = 1:-0.05:0;                 % 1.00, 0.95, ..., 0.00 (inclusive)
N = numel(x_grid);
```

part b

```
y_euler = nan(N,1);
y_euler(1) = 1.0;                   % y(1)=1
for i = 1:N-1
    yi = y_euler(i);
    % guard against division by zero / negative crossing near x=0
    if yi <= 0, yi = max(yi, 1e-12); end
    y_euler(i+1) = yi + h*f(x_grid(i), yi);
end
```

part c

```
y_rk4 = nan(N,1);
y_rk4(1) = 1.0;
for i = 1:N-1
    xi = x_grid(i); yi = y_rk4(i);
    yi = max(yi, 1e-12);
    k1 = f(xi, yi);
    k2 = f(xi + h/2, yi + (h/2)*k1);
    k3 = f(xi + h/2, yi + (h/2)*k2);
    k4 = f(xi + h,   yi + h*k3);
    y_rk4(i+1) = yi + (h/6)*(k1 + 2*k2 + 2*k3 + k4);
end
```

part d

```
x_ab4 = 1.15:-0.05:0;                     % starts at 1.15 to use seeds then march
to 0
M = numel(x_ab4);
y_ab4 = nan(M,1);

% Given seeds (align to first four entries: 1.15, 1.10, 1.05, 1.00)
seed_x = [1.15 1.10 1.05 1.00];
seed_y = [1.04768955317165, 1.03228011545637, 1.01639635681485, 1.00000000000000];
y_ab4(1:4) = seed_y(:);
```

```
% AB4 recurrence: y_{n+1} = y_n + h/24 * (55 f_n - 59 f_{n-1} + 37 f_{n-2} - 9
f_{n-3})
for n = 4:M-1
    fn   = f(x_ab4(n),   max(y_ab4(n),   1e-12));
    fn_1 = f(x_ab4(n-1), max(y_ab4(n-1), 1e-12));
    fn_2 = f(x_ab4(n-2), max(y_ab4(n-2), 1e-12));
    fn_3 = f(x_ab4(n-3), max(y_ab4(n-3), 1e-12));
    y_ab4(n+1) = y_ab4(n) + (h/24)*(55*fn - 59*fn_1 + 37*fn_2 - 9*fn_3);
end

% Extract AB4 values on the same x_grid (1.00, 0.95, ..., 0.00)
% Note: x_ab4 sequence includes 1.15 seed and then 1.10, 1.05, 1.00, 0.95, ..., 0.00
[~, ia, ib] = intersect(round(x_ab4,10), round(x_grid,10), 'stable');
y_ab4_on_grid = nan(N,1);
y_ab4_on_grid(ib) = y_ab4(ia);

%-----------------------------
% Exact values and errors
%-----------------------------
y_true = y_exact(x_grid).';
err_euler = y_true - y_euler;
err_rk4   = y_true - y_rk4;
err_ab4   = y_true - y_ab4_on_grid;

%-----------------------------
% Assemble table
%-----------------------------
T = table( ...
    x_grid.', y_true, y_euler, err_euler, y_rk4, err_rk4, y_ab4_on_grid, err_ab4,
...
    'VariableNames',
{'x','y_exact','y_euler','err_euler','y_rk4','err_rk4','y_ab4','err_ab4'} ...
);

disp('==== Q1: Exact vs Euler / RK4 / AB4 (h = 0.05 backward) ====');
```

```
==== Q1: Exact vs Euler / RK4 / AB4 (h = 0.05 backward) ====
```

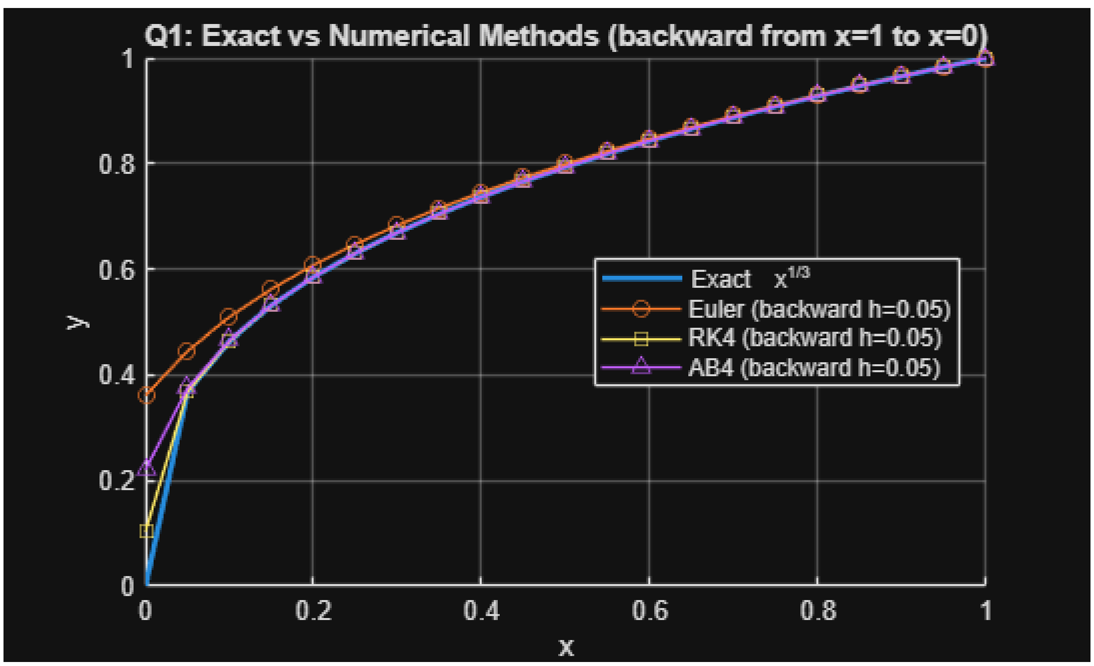
```
disp(T);
```

x	y_exact	y_euler	err_euler	y_rk4	err_rk4	y_ab4	err_ab4
1	1	1	0	1	0	1	0
0.95	0.98305	0.98333	-0.00028576	0.98305	2.3155e-10	0.98305	-3.1327e-07
0.9	0.96549	0.9661	-0.00060752	0.96549	5.3542e-10	0.96549	-7.3196e-07
0.85	0.94727	0.94824	-0.00097172	0.94727	9.3811e-10	0.94727	-1.2672e-06
0.8	0.92832	0.9297	-0.0013863	0.92832	1.478e-09	0.92832	-1.9754e-06
0.75	0.90856	0.91042	-0.0018615	0.90856	2.2119e-09	0.90856	-2.9209e-06
0.7	0.8879	0.89031	-0.00241	0.8879	3.226e-09	0.88791	-4.2001e-06
0.65	0.86624	0.86929	-0.0030487	0.86624	4.6545e-09	0.86625	-5.9595e-06
0.6	0.84343	0.84723	-0.0037994	0.84343	6.7129e-09	0.84344	-8.4267e-06
0.55	0.81932	0.82401	-0.0046917	0.81932	9.7605e-09	0.81933	-1.1966e-05

0.5	0.7937	0.79947	-0.0057665	0.7937	1.4421e-08	0.79372	-1.7181e-05
0.45	0.76631	0.77339	-0.0070812	0.76631	2.1834e-08	0.76633	-2.5119e-05
0.4	0.73681	0.74553	-0.0087198	0.73681	3.421e-08	0.73684	-3.7679e-05
0.35	0.70473	0.71554	-0.01081	0.70473	5.616e-08	0.70479	-5.8535e-05
0.3	0.66943	0.68299	-0.013555	0.66943	9.8258e-08	0.66953	-9.5319e-05
0.25	0.62996	0.64726	-0.017298	0.62996	1.8783e-07	0.63013	-0.00016547
0.2	0.5848	0.60748	-0.022672	0.5848	4.0812e-07	0.58512	-0.00031407
0.15	0.53133	0.56231	-0.030983	0.53133	1.0799e-06	0.53201	-0.00067928
0.1	0.46416	0.5096	-0.045443	0.46415	3.9945e-06	0.46597	-0.0018083
0.05	0.3684	0.44542	-0.077021	0.36837	2.8738e-05	0.37551	-0.007106
0	0	0.36142	-0.36142	0.10346	-0.10346	0.22082	-0.22082

Plot

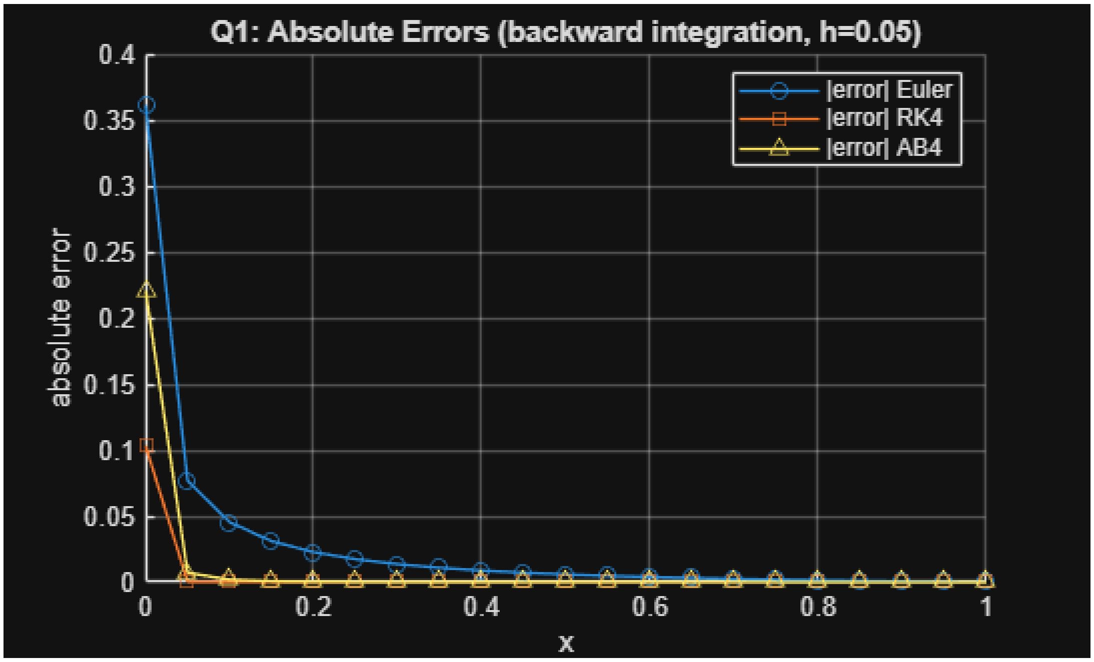
```
figure(1); clf; hold on; grid on;
plot(x_grid, y_true,  'LineWidth', 2, 'DisplayName','Exact  x^{1/3}');
plot(x_grid, y_euler, 'o-','DisplayName','Euler (backward h=0.05)');
plot(x_grid, y_rk4,   's-','DisplayName','RK4 (backward h=0.05)');
plot(x_grid, y_ab4_on_grid, '^-','DisplayName','AB4 (backward h=0.05)');
xlabel('x'); ylabel('y');
title('Q1: Exact vs Numerical Methods (backward from x=1 to x=0)');
legend('Location','best');
```




```

%------------------------------
% Plot absolute errors (semilogy to handle wide range)
%------------------------------
figure(2); clf; hold on; grid on;
semilogy(x_grid, abs(err_euler), 'o-','DisplayName','|error| Euler');
semilogy(x_grid, abs(err_rk4),   's-','DisplayName','|error| RK4');
semilogy(x_grid, abs(err_ab4),   '^-','DisplayName','|error| AB4');
xlabel('x'); ylabel('absolute error');
```

```
title('Q1: Absolute Errors (backward integration, h=0.05)');
legend('Location','best');
```




```
%------------------------------
% Brief accuracy summary in Command Window
%------------------------------
fprintf('\nMax |error| (Euler): %g\n', max(abs(err_euler),[],'omitnan'));
```

```
Max |error| (Euler): 0.361419
```

```
fprintf('Max |error| (RK4)   : %g\n', max(abs(err_rk4),[],'omitnan'));
```

```
Max |error| (RK4)   : 0.103455
```

```
fprintf('Max |error| (AB4)   : %g\n\n', max(abs(err_ab4),[],'omitnan'));
```

```
Max |error| (AB4)   : 0.220816
```